

Data Privacy: Midterm Solutions

1. For $f : \mathcal{X}^n \rightarrow \mathbb{R}$ with ℓ_1 sensitivity Δ , the truncated Laplacian mechanism is

$$\mathcal{M}(X) = f(X) + Z$$

where Z is $\text{LapTr}(\lambda, \tau)$. Recall that the Truncated Laplacian random variable has pdf

$$f_Z(z) = \begin{cases} C \exp(-|z|/\lambda) & \text{if } |z| \leq \tau \\ 0 & \text{if } |z| > \tau \end{cases}$$

where C is a normalization constant.

- (a) Find C . (5 points)
(b) Show that the Truncated Laplacian mechanism is (ε, δ) -DP. You may assume $\lambda = \Delta/\varepsilon$ and $\tau < \Delta$, but you need to specify δ . (10 points)

Solution.

- (a) Note that

$$\begin{aligned} \int_{-\tau}^{\tau} \exp(-|z|/\lambda) dz &= 2 \int_0^{\tau} \exp(-z/\lambda) dz \\ &= 2\lambda(1 - \exp(-\tau/\lambda)). \end{aligned}$$

Thus,

$$C = \frac{1}{2\lambda(1 - \exp(-\tau/\lambda))}.$$

- (b) Let $\lambda = \Delta/\varepsilon$. For y such that $|f(X) - y| \leq \tau$ and $|f(X') - y| \leq \tau$, the privacy loss random variable

$$\begin{aligned} I_{X, X'}(y) &= \log \frac{\Pr[\mathcal{M}(X) = y]}{\Pr[\mathcal{M}(X') = y]} \\ &= -\frac{|f(X) - y|}{\lambda} + \frac{|f(X') - y|}{\lambda} \\ &\leq \frac{|f(X) - f(X')|}{\lambda} \\ &\leq \frac{\Delta}{\lambda} \\ &= \varepsilon. \end{aligned}$$

Now, consider the probability of bad events

$$\begin{aligned}
\Pr[|f(X) - y| > \tau \text{ or } |f(X') - y| > \tau] &= \Pr[|Z| > \tau \text{ or } |f(X') - f(X) - Z| > \tau] \\
&\leq \Pr[|\Delta - Z| > \tau] \\
&= \Pr[Z > \tau + \Delta \text{ or } Z < \Delta - \tau] \\
&= \int_{-\tau}^{\Delta - \tau} C \exp(-|z|/\lambda) dz \\
&= -C\lambda (\exp(-\tau/\lambda) - \exp(-(\tau - \Delta)/\lambda)) \\
&= -C\lambda \exp(-\tau/\lambda)(1 - \exp(\Delta/\lambda)) \\
&= \frac{-\exp(-\tau/\lambda)(1 - \exp(\Delta/\lambda))}{2(1 - \exp(-\tau/\lambda))}.
\end{aligned}$$

Thus, it is (ε, δ) -DP when

$$\delta = \frac{\exp(-\tau\varepsilon/\Delta)(\exp(\varepsilon) - 1)}{2(1 - \exp(-\tau\varepsilon/\Delta))}.$$

2. Let $S = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$. Let $X = (X_1, \dots, X_n)$ where $X_i \in S$ for all $1 \leq i \leq n$.

- (a) Consider a mechanism $M_S : S^n \rightarrow S$ that returns $M_S(X) = X_I$ for where $I \sim \text{Unif}\{1, \dots, n\}$. Show that M_S cannot be a $(\varepsilon, 0)$ -DP. (5 points)
- (b) Show that M_S is $(0, \delta)$ -DP. Specify δ . (10 points)
- (c) Suppose the utility that you are considering is to approximate the empirical mean μ , where $\mu = \frac{1}{n} \sum_{i=1}^n X_i$. Show that

$$\mathbb{E}[\|M_S(X) - \mu\|^2] = 1 - \|\mu\|^2.$$

(10 points)

Solution.

- (a) Consider $X = ((1, 0), (0, 1), (0, 1), \dots, (0, 1))$ and $X' = ((0, 1), (0, 1), (0, 1), \dots, (0, 1))$. Probability of $\Pr[M_S(X) = (1, 0)] = 1/n$ but $\Pr[M_S(X') = (1, 0)] = 0$.
- (b) It is $(0, 1/n)$ -DP.
- (c) We have

$$\begin{aligned}
\mathbb{E}[\|M_S(X) - \mu\|^2] &= \sum_{j=1}^n \frac{1}{n} \|X_j - \mu\|^2 \\
&= \sum_{j=1}^n \frac{1}{n} (\|X_j\|^2 - 2\langle X_j, \mu \rangle + \|\mu\|^2) \\
&= 1 - \sum_{j=1}^n \frac{1}{n} 2\langle \mu, \mu \rangle + \|\mu\|^2 \\
&= 1 - \|\mu\|^2.
\end{aligned}$$

3. Suppose $M : \{0, 1\}^n \rightarrow \mathcal{Y}$ is an (ε, δ) -differentially private mechanism, and let $A : \mathcal{Y} \rightarrow \{0, 1\}^n$ be an adversary attempting to reconstruct the sensitive bits in a dataset $X \in \{0, 1\}^n$ from the output $M(X)$. More precisely, given $M(X) = Y$, the adversary generates $A(Y) = \hat{X} = (\hat{X}_1, \dots, \hat{X}_n)$, and wants to maximize the fraction of correct guesses

$$\frac{1}{n} \sum_{i=1}^n \mathbf{1}(X_i = \hat{X}_i)$$

where

$$\mathbf{1}(X_i = \hat{X}_i) = \begin{cases} 1 & \text{if } X_i = \hat{X}_i \\ 0 & \text{if } X_i \neq \hat{X}_i. \end{cases}$$

Assume the dataset is a random variable $X = (X_1, \dots, X_n)$ consisting of n independent draws from a Bernoulli(p) distribution, for a known value of $p < 1/2$.

- (a) Let $Z_i = \mathbf{1}(X_i = \hat{X}_i)$. Show that

$$\mathbb{E}[Z_i] = \Pr[\hat{X}_i = 1 \mid X_i = 1]p + \Pr[\hat{X}_i = 0 \mid X_i = 0](1 - p)$$

(5 points)

- (b) Since M is (ε, δ) -DP, for any neighboring dataset $X \sim X'$, we have

$$\Pr[\hat{X}_i = 1 \mid X] \leq e^\varepsilon \Pr[\hat{X}_i = 1 \mid X'] + \delta$$

Then, show that

$$\Pr[\hat{X}_i = 1 \mid X_i = 1] \leq e^\varepsilon \Pr[\hat{X}_i = 1 \mid X_i = 0] + \delta$$

(10 points)

- (c) Prove that the expected fraction of bits that the adversary successfully reconstructs is not much larger than the trivial bound of $1 - p$, which could be achieved by always guessing the all-zero dataset. Specifically, show that:

$$\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n Z_i\right] \leq e^\varepsilon \cdot (1 - p) + \delta.$$

(10 points)

Solution.

- (a) We have

$$\begin{aligned} \mathbb{E}[Z_i] &= \Pr[Z_i = 1] \\ &= \Pr[\hat{X}_i = X_i] \\ &= \Pr[\hat{X}_i = X_i \mid X_i = 1] \Pr[X_i = 1] + \Pr[\hat{X}_i = X_i \mid X_i = 0] \Pr[X_i = 0] \\ &= \Pr[\hat{X}_i = 1 \mid X_i = 1]p + \Pr[\hat{X}_i = 0 \mid X_i = 0](1 - p). \end{aligned}$$

(b)

$$\begin{aligned}
& \Pr[\hat{X}_i = 1 \mid X_i = 1] \\
&= \sum_{X: X_i=1} \Pr[X = (X_1, \dots, X_{i-1}, 1, X_{i+1}, \dots, X_n) \mid X_i = 1] \Pr[\hat{X}_i = 1 \mid X] \\
&\leq \sum_{X: X_i=1} \Pr[X = (X_1, \dots, X_{i-1}, 0, X_{i+1}, \dots, X_n) \mid X_i = 0] \Pr[\hat{X}_i = 1 \mid X] \\
&\leq \sum_{X': X'_i=0} \Pr[X' = (X'_1, \dots, 0, \dots, X'_n)] (e^\varepsilon \Pr[\hat{X}_i = 1 \mid X'] + \delta) \\
&= e^\varepsilon \Pr[\hat{X}_i = 1 \mid X_i = 0] + \delta.
\end{aligned}$$

(c) Similar from part (b), we have

$$\Pr[\hat{X}_i = 0 \mid X_i = 0] \leq e^\varepsilon (1 - \Pr[\hat{X}_i = 1 \mid X_i = 1]) + \delta.$$

Thus,

$$\begin{aligned}
\mathbb{E}[Z_i] &= \Pr[\hat{X}_i = 1 \mid X_i = 1]p + \Pr[\hat{X}_i = 0 \mid X_i = 0](1-p) \\
&\leq \Pr[\hat{X}_i = 1 \mid X_i = 1]p + (1-p)(e^\varepsilon - e^\varepsilon \Pr[\hat{X}_i = 1 \mid X_i = 1]) + (1-p)\delta \\
&= (1-p)e^\varepsilon + (p - e^\varepsilon(1-p)) \Pr[\hat{X}_i = 1 \mid X_i = 1] + (1-p)\delta \\
&\leq (1-p)e^\varepsilon + \delta.
\end{aligned}$$

Then, the average is clearly bounded by the same:

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n Z_i \right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[Z_i] \leq (1-p)e^\varepsilon + \delta.$$

4. Suppose you are given a private dataset $X \in \mathcal{X}^n$, where $\mathcal{X} = \{1, \dots, d\}$. That is, the dataset X consists of n integers $x_1, \dots, x_n$ between 1 and d . We want to generate a sample $Y \in \{1, \dots, d\}$ that satisfies $\min_{i=1}^n x_i \leq Y \leq \max_{i=1}^n x_i$ with high probability, but of course with ε -DP guarantee.

(a) Define a utility function $q : \mathcal{X}^n \times \mathcal{X} \rightarrow \mathbb{Z}$ as:

$$q(X, y) = \min \{ |\{i : x_i \leq y\}|, |\{i : x_i \geq y\}| \}$$

where $|\{\cdot\}|$ denotes the size of the set. Let $\text{OPT}(X) = \max_{y \in \mathcal{X}} q(X, y)$, then show that

$$\text{OPT}(X) \geq \frac{n-1}{2}.$$

(5 points)

- (b) We use the exponential mechanism with range $\mathcal{X}$ with utility function q . Find the sensitivity Δ . (10 points)
- (c) Describe an exponential mechanism that satisfies ε -differential privacy, and show that

$$\Pr \left[q(X, Y) \geq \text{OPT}(X) - \frac{2}{\varepsilon} \log \left(\frac{d}{\beta} \right) \right] \geq 1 - \beta.$$

You can use the utility guarantee result from the lecture note. (10 points)

(d) Specify C_1 and C_2 that the above exponential mechanism satisfies

$$\Pr \left[\min_{i=1}^n x_i \leq y \leq \max_{i=1}^n x_i \right] \geq 1 - \beta$$

for

$$n \geq \frac{C_1}{\varepsilon} \ln \left(\frac{d}{\beta} \right) + C_2.$$

Here, C_1 and C_2 should be constants that are independent of n , d , β , and ε .
(Hint) Consider the event $q(X, y) \geq 1$. (10 points)

Solution.

(a) Since we can choose y to be the $\lfloor n/2 \rfloor$ -th element of the sorted list, we have

$$\text{OPT}(X) \geq \left\lfloor \frac{n}{2} \right\rfloor \geq \frac{n-1}{2}.$$

(b) We now compute the sensitivity Δ of the utility function q . Let X and X' be neighboring datasets differing in one element. Then for any $y \in \mathcal{X}$,

$$\begin{aligned} ||\{i : x_i \leq y\}| - |\{i : x'_i \leq y\}|| &\leq 1, \\ ||\{i : x_i \geq y\}| - |\{i : x'_i \geq y\}|| &\leq 1. \end{aligned}$$

Thus,

$$|q(X, y) - q(X', y)| \leq 1,$$

so $\Delta = 1$.

(c) Check the lecture note. We have $\Delta = 1$, $t = \log(1/\beta)$.

(d) Observe that y lies between $\min_i x_i$ and $\max_i x_i$ if and only if $q(X, y) \geq 1$.
From part (c), with probability at least $1 - \beta$, the output y satisfies:

$$q(X, y) \geq \text{OPT}(X) - \frac{2}{\varepsilon} \ln \left(\frac{|\mathcal{X}|}{\beta} \right).$$

As noted earlier, $\text{OPT}(X) \geq \frac{n-1}{2}$, so if

$$\frac{n-1}{2} - \frac{2}{\varepsilon} \ln \left(\frac{d}{\beta} \right) \geq 1,$$

then $q(X, y) \geq 1$ and therefore y lies between $\min_i x_i$ and $\max_i x_i$.
Solving the inequality,

$$\frac{n-1}{2} \geq \frac{2}{\varepsilon} \ln \left(\frac{d}{\beta} \right) + 1 \quad \Rightarrow \quad n \geq \frac{4}{\varepsilon} \ln \left(\frac{d}{\beta} \right) + 3.$$

Hence, the desired guarantee holds if we set constants $C_1 = 4$ and $C_2 = 3$:

$$n \geq \frac{4}{\varepsilon} \ln \left(\frac{d}{\beta} \right) + 3 \quad \Rightarrow \quad \Pr \left[\min_i x_i \leq y \leq \max_i x_i \right] \geq 1 - \beta.$$